A Möbius-binomial C/F bridge and a claim-disciplined theorem-transfer atlas for Apéry-like sources

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Abstract

We isolate a direct Möbius-binomial bridge between Zagier's C and F Apéry-like sequences:

$$F(x) = (1-9x)^{-1} C\left(\frac{-x}{1-9x}\right).$$

Equivalently, the bridge is an involutive binomial coefficient transform between the two catalog-normalized coefficient sequences and preserves their ordinary Hankel determinants and, by integral unimodular congruence, their Smith forms. It also transports the differential equation of C to that of F and is cross-checked by the known Franel transform route. We then use this bridge as a primitive for a theorem-transfer atlas. The atlas distinguishes results proved in this paper, theorem imports by reference, conditional transfers, finite exact replays, source-attached conjectures, proof obligations, blocked rows, bounded negatives, watch metadata, and closed metadata. In particular, we do not derive Sun's Conjecture 3.5 from the C/F bridge, do not promote finite replays or coefficient matches to theorems, and do not turn Dwork or Rowland–Yassawi metadata into unconditional theorem claims. Sun's raw-C conjecture is the subject of a separate supercongruence manuscript [Bab26]. The result here is a claim-disciplined package for using a proved C/F bridge while keeping adjacent Apéry-like source layers separate.

1 Introduction

Apéry-like sequences sit at the intersection of several structures: recurrences, modular parametrizations, binomial transforms, constant-term and Laurent-polynomial models, supercongruences, and automatic congruences. These structures often appear in clusters, but they do not transfer automatically from one sequence to another. A coefficient match, a shared recurrence shape, or a common modular level may suggest a relation; none of these is by itself a theorem-transfer mechanism.

This point of view is close in spirit to the tabular literature around Calabi–Yau and Apéry-like differential equations, where large lists of operators, transformations, and periods are useful only when their normalizations and equivalences are explicit [AvEvSZ05, AvSZ11]. Our scope is much narrower: a claim-disciplined atlas centered on one proved C/F bridge.

This paper is organized around one case where the transfer mechanism is explicit. We prove and certify a direct Möbius-binomial bridge between the Zagier C and F sequences:

$$F(x) = (1-9x)^{-1} C\left(\frac{-x}{1-9x}\right).$$

The identity gives an involutive coefficient transform, an ODE pullback, and a Hankel determinant invariance, together with a constant-term complement operation. It is compatible with the known Franel network around these sequences.

The second contribution is methodological. We use the C/F bridge as a primitive in a theorem-transfer atlas for nearby Apéry-like source layers. The atlas records what can be transferred, what can only be imported by reference, what is merely metadata, and what remains

a proof obligation. The point is not to claim that all adjacent congruence frameworks become theorems of this paper. Rather, we separate proof, imported theorem, finite replay, conjectural source attachment, blocked metadata, watch rows, and bounded negative evidence.

The bridge is the mathematical core; the atlas records how far its transport consequences extend. Sections 2–4 give the C/F bridge and its certificate. Sections 5–11 explain how the bridge is used to build a disciplined theorem-transfer atlas. Appendices contain the theta-gauge certificate, the Atkin–Lehner normalization audit, the constant-term complement guardrail, selected registry rows, open problems, and source-status notes.

2 Zagier C and F

Let C_n and F_n denote Zagier's C and F Apéry-like sequences in the catalog convention used throughout this paper [Zag09, OEI26a, OEI26b]. Write

$$C(x) = \sum_{n\geq 0} C_n x^n, \qquad F(x) = \sum_{n\geq 0} F_n x^n.$$

The C sequence is given by

$$C_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{2k}{k}.$$

It satisfies

$$(n+1)^2 C_{n+1} = (10n^2+10n+3)C_n - 9n^2 C_{n-1}.$$

The F sequence is normalized by $F_0 = 1$, $F_1 = 6$, and satisfies

$$(n+1)^2 F_{n+1} = (17n^2+17n+6)F_n - 72n^2 F_{n-1}.$$

The atlas below always treats these catalog-normalized objects as the named source and target rows.

3 The C/F Möbius-binomial bridge

3.1 Modular parameter and theta-gauge normalization

Write $\eta_d(\tau) = \eta(d\tau)$ and $q = e^{2\pi i\tau}$. We use the modular parametrizations in Beukers, Sections 8.7 and 8.10 [Beu25]:

$$t_C = \left(\frac{\eta_6^2\eta_1}{\eta_2^2\eta_3}\right)^4, \qquad h = \frac{\eta_6^5\eta_2}{\eta_1^5\eta_3}, \qquad t_C = \frac{h}{1+9h}.$$

The associated weight-one forms are

$$\Theta_C = \frac{\eta_2^6\eta_3}{\eta_1^3\eta_6^2} = C(t_C), \qquad \Theta_F = \frac{\eta_1^6\eta_6}{\eta_2^3\eta_3^2} = F(-h).$$

The F row in the source is twisted by $(-1)^n$. Thus the source parameter is $t_F^{\mathrm{source}} = h$ and the catalog parameter is

$$t_F = t_F^{\mathrm{cat}} = -h = -\frac{t_C}{1-9t_C}.$$

These weight-zero parameters are meromorphic modular functions on $X_0(6)$. The two theta forms have the same character, so their ratio also has weight zero and trivial character. The eta products have no zeros in the upper half-plane, and their possible zeros and poles are cuspidal. At infinity,

$$t_C = q + O(q^2), \qquad h = q + O(q^2), \qquad t_F^{\mathrm{cat}} = -q + O(q^2).$$

In particular the catalog F parameter is an invertible formal change of variable. We also have

$$1 + 9t_F^{\mathrm{source}} = 1 - 9t_F^{\mathrm{cat}} = \frac{1}{1 - 9t_C}.$$

The theta-ratio identity needed below is proved by the explicit eta-product valence certificate in Appendix A.

Theorem 3.1 (Zagier C/F Möbius-binomial bridge)**.** *Under the catalog convention used here and the verified F twist transport,*

$$F(x) = (1 - 9x)^{-1} C\left(\frac{-x}{1 - 9x}\right).$$

Proof. With the normalization of Section 3.1, the theta-gauge identities are

$$\Theta_F/\Theta_C = 1 - 9t_C, \qquad \Theta_C/\Theta_F = 1 + 9t_F^{\mathrm{source}} = 1 - 9t_F^{\mathrm{cat}}.$$

The second form is the one compatible with the catalog parameter. With

$$u = t_C(q), \qquad x = t_F^{\mathrm{cat}}(q) = \frac{-u}{1 - 9u},$$

one has

$$\frac{-x}{1 - 9x} = u, \qquad (1 - 9x)^{-1} = 1 - 9u.$$

Thus

$$F(x) = \Theta_F(q) = (1 - 9u)\Theta_C(q) = (1 - 9x)^{-1} C\left(\frac{-x}{1 - 9x}\right).$$

The forward theta-ratio identity is proved by the eta-product valence certificate in Appendix A; the inverse identity follows algebraically from it and the parameter relation above. Since $x = t_F^{\mathrm{cat}}(q) = -q + O(q^2)$ is an invertible formal change of variable, the resulting identity after substitution is an identity of generating series in x. Appendix A.2 gives a separate Atkin–Lehner W_2 normalization audit, which is not a substitute for this bridge proof.

□

Corollary 3.2 (Coefficient transform)**.** *Define*

$$T(a)_m = \sum_{n=0}^{m} \binom{m}{n} (-1)^n 9^{m-n} a_n.$$

Then $T(C) = F$ and $T(F) = C$.

Proof. Taking coefficients in Theorem 3.1 gives $T(C) = F$. Let $g(x) = -x/(1 - 9x)$. Then $g(g(x)) = x$ and

$$(1 - 9x)^{-1}(1 - 9g(x))^{-1} = 1.$$

Applying the same transform a second time therefore gives $T(F) = C$. □

Corollary 3.3 (Hankel determinants)**.** *For a sequence $A = (A_n)_{n \geq 0}$ put*

$$\Delta_N(A) = \det(A_{i+j})_{0 \leq i,j < N}.$$

Then, for every $N \geq 1$,

$$\Delta_N(C) = \Delta_N(F).$$

Proof. This is the standard Hankel invariance of binomial transforms [SS06], but the short argument is included to fix the signs. Set $G_m = (-1)^m F_m$. By Corollary 3.2,

$$G_m = \sum_{n=0}^{m} \binom{m}{n} (-9)^{m-n} C_n.$$

For fixed N, let B_N be the lower triangular matrix

$$(B_N)_{i,j} = \begin{cases} \binom{i}{j}(-9)^{i-j}, & j \le i, \\ 0, & j > i, \end{cases} \qquad 0 \le i, j < N.$$

The matrix B_N has determinant 1, and Vandermonde convolution gives

$$(G_{i+j})_{0\le i,j<N} = B_N \, (C_{i+j})_{0\le i,j<N} \, B_N^t.$$

Hence $\Delta_N(G) = \Delta_N(C)$. Finally,

$$(F_{i+j})_{0\le i,j<N} = D_N \, (G_{i+j})_{0\le i,j<N} \, D_N, \qquad D_N = \operatorname{diag}((-1)^0, \ldots, (-1)^{N-1}),$$

so $\Delta_N(F) = \Delta_N(G)$. □

Corollary 3.4 (Integral Hankel congruence and Smith transport)**.** *For every $N \ge 1$, the ordinary Hankel matrices $H_N(A) = (A_{i+j})_{0\le i,j<N}$ satisfy*

$$H_N(F) = U_N H_N(C) U_N^t, \qquad (U_N)_{ij} = (-1)^i \binom{i}{j} (-9)^{i-j} \quad (j \le i),$$

with zero entries for $j > i$ and $U_N \in \mathrm{GL}_N(\mathbb{Z})$. Consequently they have the same integral Smith normal form, including their rank and zero invariant factors, and the same local Smith form at every prime.

Proof. The proof of Corollary 3.3 gives $U_N = D_N B_N$. The integer matrix B_N is unitriangular (its inverse replaces -9 by 9), and D_N has diagonal entries ± 1. Thus U_N and its transpose are unimodular. Left and right multiplication by such matrices preserve the integral Smith normal form. □

This is an equality of Smith forms, not a classification of their invariant factors. It applies to ordinary Hankel matrices; for example the shifted order-one matrices $(C_1) = (3)$ and $(F_1) = (6)$ have different Smith forms.

Corollary 3.5 (ODE pullback)**.** *Let*

$$L_C = x(x-1)(9x-1)D_x^2 + (27x^2 - 20x + 1)D_x + 3(3x-1)$$

be the differential operator associated to the C recurrence, and let

$$L_F = x(72x^2 - 17x + 1)D_x^2 + (216x^2 - 34x + 1)D_x + 6(12x-1)$$

be the corresponding F operator. Under

$$u = -\frac{x}{1-9x}, \qquad F(x) = (1-9x)^{-1} C(u),$$

symbolic substitution gives

$$L_C[C](u) = (9x-1) L_F[F](x).$$

Thus the C equation pulls back to the F equation up to the rational factor $9x-1$, which is nonzero as a rational function.

Remark 3.6 (Franel route)**.** *Let $R_n = \sum_{k=0}^{n} \binom{n}{k}^3$. The known Franel transforms are*

$$C(x) = (1-x)^{-1} R\left(\frac{x}{1-x}\right), \qquad F(x) = (1-8x)^{-1} R\left(\frac{-x}{1-8x}\right),$$

with provenance through Barrucand, Callan, Verrill, and OEIS [Bar75, Cal08, Ver10]. Composing these formulas gives the same C/F bridge.

4 Certificate and normalization

The proof uses the explicit common $\Gamma_0(6)$ eta-product normalization, the source-to-catalog F sign twist, and a theta-ratio valence certificate. The certificate lists the possible cusp poles and derives the sufficient vanishing at infinity directly from the eta products. The inverse ratio identity is an algebraic consequence. No large q-expansion replay is needed for this proof; the details are in Appendix A.

5 Additional functorial consequences

The C/F bridge is used functorially in a small number of controlled ways. The consequences proved in this paper are the coefficient involution, the Hankel determinant invariance, the ODE pullback, and the binomial-complement constant-term lemma. Together with the imported C-side Laurent model, that lemma gives the $9 - \Lambda_{C,1}$ complement witness. The source-layer consequences below explain how those facts are recorded in the atlas. They do not promote adjacent Rowland–Yassawi, BTY, or Dwork comparison metadata into unconditional theorem claims.

5.1 Coefficient transform and Liu normalization

The coefficient transform in Corollary 3.2 is the atlas primitive. It allows a theorem stated for F to be rewritten as a theorem for $T(C)$. It does not automatically produce a theorem for raw C_n.

In Liu's normalization

$$v_m^{(A)}(u) = \sum_{n=0}^{m} \binom{m}{n} (-u)^{m-n} A_n$$

[Liu25], the same bridge reads $v_m^{(C)}(9) = (-1)^m F_m$ and, by involution, $v_m^{(F)}(9) = (-1)^m C_m$. This is a normalization map into Liu's binomial-transform layer, not a blanket import of every adjacent congruence.

For example, Sun's source rows include congruences for a sequence Q_n with $Q_n = (-1)^n F_n$ in the relevant convention [Sun22]. The bridge rewrites these as statements about $(-1)^n T(C)_n$. Sun's raw-C conjectural congruence is not a consequence of this C/F bridge. A separate manuscript reports its resolution [Bab26].

5.2 Franel network

Remark 3.6 explains why the bridge is not an isolated accident: C and F are both connected to the Franel sequence by known Möbius-binomial transforms. The theorem-transfer atlas records this as known source metadata. It does not treat every inverse or coefficient replay as a new theorem edge.

The source-layer order is important: the Franel sequence is the common upstream object, while the C/F bridge is the induced relation after the two classical transforms are composed. This is why the paper can use Franel as a provenance check without making every Franel-side theorem a C/F theorem.

More concretely, both generating functions are obtained from the same cubic Franel series $R(x) = \sum R_n x^n$ by two different fractional-linear changes of variable. The C/F bridge is therefore the composition of two classical Franel transforms, not a relation inferred only from matching initial coefficients. This Franel viewpoint is useful in the atlas because it separates a known structural network from later theorem-transfer claims.

5.3 Constant-term complement

The coefficient transform has a direct constant-term interpretation.

Lemma 5.1 (Binomial-complement constant-term transform)**.** *Let Λ be a Laurent polynomial and let*

$$A_n = \mathrm{CT}(\Lambda^n).$$

For a scalar a, define

$$B_n = \sum_{k=0}^{n} \binom{n}{k} (-1)^k a^{n-k} A_k.$$

Then

$$B_n = \mathrm{CT}((a - \Lambda)^n).$$

Proof. By the binomial theorem,

$$(a - \Lambda)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} (-\Lambda)^k.$$

Taking constant terms gives

$$\mathrm{CT}((a - \Lambda)^n) = \sum_{k=0}^{n} \binom{n}{k} (-1)^k a^{n-k} \,\mathrm{CT}(\Lambda^k) = B_n.$$

□

For

$$\Lambda_{C,1} := \Lambda_C = \frac{(x + y + 1)(xy + x + y)}{xy}$$

Gorodetsky records this Laurent model for the C row [Gor23]. Equivalently, if $P = (x + y + 1)(xy + x + y)$, then

$$\mathrm{CT}(\Lambda_{C,1}^n) = [x^n y^n] P^n,$$

and the cited model identifies this coefficient with the catalog-normalized C_n. Lemma 5.1 gives

$$F_n = \mathrm{CT}((9 - \Lambda_{C,1})^n).$$

This is a constant-term operation. It is not a Laurent mutation claim and it is not a Newton-equivalence claim between $9 - \Lambda_{C,1}$ and other F-side models such as Gorodetsky's replacement model Q_F [Gor23].

5.4 Rational diagonals and small automata

The constant-term models also admit rational diagonal contracts, which support small automatic-congruence metadata for C/F and related rows. This paper records those automata as metadata. It does not distribute the complete state constructions behind the recorded fixed-modulus automata. Rowland–Yassawi's Theorem 2.1 requires polynomial numerator and denominator over $\mathbb{Z}_p$ with denominator constant term a unit; for an explicit rational diagonal there is no additional generic singularity filter. Producing and verifying the relevant Cartier states is a separate obligation [RY15].

6 Claim classes and atlas discipline

The atlas uses explicit claim classes. Table 1 summarizes the ones used in this paper.

The guardrails are simple: finite replay is evidence, not proof; coefficient matching is not theorem transfer; and external theorems remain external theorems.

`proved_in_paper`	Proved in the current paper.
`theorem_import_by_reference`	External theorem used with its source label; not presented as a theorem of this paper.
`conditional_theorem_transfer`	Transfer valid after an explicit bridge or source assumption.
`closed_metadata`	Verified auxiliary record, such as a normalization map, fixed-modulus automaton, source-label reconciliation, or known literature identity; not used as a new theorem unless separately promoted through a proof route.
`explicit_automaton_metadata`	Explicit finite-state or prime-power automaton data recorded as metadata; in the V2 package this is treated as a closed-metadata subtype, not as a theorem-transfer class.
`finite_exact_replay`	Exact finite check used as evidence only.
`source_attached_conjecture`	Conjectural source row attached to the atlas without proof.
`proof_obligation`	A precise missing lemma, not a vague direction.
`blocked`	A known missing contract prevents theorem use.
`watch`	Interesting metadata without a promotion route.
`bounded_negative`	A bounded search failed a stated gate.

Table 1: Claim classes used in the theorem-transfer atlas.

7 The theorem-transfer registry

The registry is an operational table of theorem imports and bridge-generated transfers. The complete table is part of the data package; Appendix C records the schema and selected rows. The core examples are:

1. the C/F bridge, proved in this paper;
2. the coefficient involution, proved in this paper;
3. the ODE pullback, proved in this paper;
4. Sun's F-side source rows transferred only to $T(C)$;
5. Sun's raw-C target, retained in the historical V1/V2 atlas as a source-attached conjecture and proof obligation, but now marked in the paper-facing readout as addressed separately by [Bab26];
6. CT/Newton guardrails, which block mutation or Newton-equivalence claims without an explicit operation.

8 Source-layer examples

8.1 Sun layer

Sun supplies source congruences for F-side objects and related binomial sums [Sun22]. The C/F bridge transfers F-side statements to $T(C)$. It does not prove raw-C variants from C/F data. The raw-C Conjecture 3.5 is the subject of a separate manuscript reporting a proof [Bab26], and the atlas keeps that proof route separate from the C/F theorem-transfer rows.

8.2 Dwork layer

Mellit–Vlasenko give Dwork-type congruences for constant terms of powers of Laurent polynomials [MV16]. The complement model $9 - \Lambda_{C,1}$ gives an additional F-side constant-term witness. We record it as Dwork metadata by reference. We do not claim a Newton equivalence or a Laurent mutation between this complement model and the F-side model Q_F.

Thus the atlas keeps three objects separate: the C-side Laurent model, the bridge-generated complement $9 - \Lambda_{C,1}$, and independent F-side Laurent models such as Q_F. The first two give a closed C/F complement witness; comparing the complement to other F-side models remains open.

8.3 Automatic congruence layer

Rowland–Yassawi provide the source framework for automatic congruences of rational diagonals [RY15]. The atlas records explicit small automata for Cooper, Franel, and C/F small moduli, including C/F mod 2, mod 3, mod 4, and mod 9 metadata. These are explicit automaton or prime-power metadata; they are not a full Rowland–Yassawi theorem import.

8.4 Chan–Cooper companion Lucas lane

The BTY/Chan–Cooper audit separates the raw weight-one problem from a weight-two Chan–Cooper companion layer. The raw weight-one $\chi(w_d)$ values remain uncomputed. However, the five companion rows V6A, V6B, V6C, V8, and V9 admit direct Chan–Cooper binomial formulas [CC12], and their companion coefficient sequences satisfy Lucas congruences modulo every prime by a direct Lucas–Kummer argument. This result is not a C/F bridge transfer and is not a BTY Theorem 1.1 import.

Let

$$U_i(n) = \binom{2n}{n} s_i(n).$$

The source formulas (Chan–Cooper, Table 2) are

$$s_{6A}(n) = \sum_{j=0}^{n} (-8)^{n-j} \binom{n}{j} \sum_{\ell=0}^{j} \binom{j}{\ell}^3,$$

$$s_{6B}(n) = \sum_{j=0}^{n} \binom{n}{j}^2 \binom{2j}{j}, \qquad s_{6C}(n) = \sum_{j=0}^{n} \binom{n}{j}^3,$$

$$s_8(n) = \sum_{j} 4^{n-2j} \binom{n}{2j} \binom{2j}{j}^2,$$

and

$$s_9(n) = \sum_{j} (-3)^{n-3j} \binom{n}{j} \binom{n-j}{j} \binom{n-2j}{j}.$$

The sums are taken over the natural ranges where the displayed binomial coefficients are defined. The Chan–Cooper table gives the positive-q coefficient sequence U_8; the atlas-normalized local V8 branch uses $Z_8(-q)$, hence the coefficient sequence $U_8^-(n) = (-1)^n U_8(n)$. We record both because Lucas congruences are preserved by this sign twist.

Lemma 8.1 (Lucas summation closure)**.** *Let $S(n) = \sum_{j \geq 0} f(n, j)$ be a finite integer-valued sum, with $f(n, j) = 0$ outside its natural support. Fix a prime p. Suppose that, for all $N, J \geq 0$ and $0 \leq n_0, j_0 < p$, including terms outside the support,*

$$f(n_0 + pN, j_0 + pJ) \equiv f(n_0, j_0) f(N, J) \pmod{p}.$$

Then $S(n_0 + pN) \equiv S(n_0)S(N) \pmod{p}$. If this hypothesis holds for every prime, S satisfies Lucas congruences modulo every prime.

Proof. Partition the summation index as $j = j_0 + pJ$ and factor the resulting finite double sum modulo p using the stated congruence. □

Theorem 8.2 (Chan–Cooper companion Lucas rows)**.** *The five sequences*

$$U_{6A}, U_{6B}, U_{6C}, U_8, U_9$$

satisfy Lucas congruences modulo every prime. The negative-q level-8 sequence

$$U_8^-(n) = (-1)^n U_8(n)$$

also satisfies Lucas congruences modulo every prime.

Proof. We use the standard Lucas theorem for binomial and multinomial coefficients, together with the Kummer carry criterion and Lemma 8.1.

First, the sequence a^n satisfies Lucas congruences modulo every prime. If $p \nmid a$, this follows from Fermat's theorem; if $p \mid a$, the sequence is congruent to $1, 0, 0, \ldots$ modulo p. Products of Lucas sequences are Lucas, and the central binomial sequence $\binom{2n}{n}$ is Lucas, with digit-doubling carries producing zero factors.

The Franel cubic sum

$$R(n) = \sum_j \binom{n}{j}^3$$

is Lucas by applying Lucas theorem termwise. This proves the claim for s_{6C}. The same argument, with the central binomial factor $\binom{2j}{j}$, proves the claim for

$$s_{6B}(n) = \sum_j \binom{n}{j}^2 \binom{2j}{j}.$$

For s_{6A}, write

$$s_{6A}(n) = \sum_j \binom{n}{j} R(j)(-8)^{n-j}.$$

This is a binomial convolution of Lucas sequences, hence is Lucas.

For s_8, when p is odd, Lucas and Kummer factor the summand

$$4^{n-2j} \binom{n}{2j} \binom{2j}{j}^2$$

digitwise; in carry cases both the large summand and the low/high product vanish modulo p. For $p = 2$, one has $s_8(0) = 1$ and $s_8(n) \equiv 0 \pmod{2}$ for $n > 0$, since the $j = 0$ term contains 4^n and $\binom{2j}{j}$ is even for $j > 0$.

For s_9, when $p \neq 3$, the multinomial form

$$(-3)^{n-3j} \frac{n!}{j!j!j!(n-3j)!}$$

factors digitwise in carry-free cases; in carry cases both the large summand and the low/high product vanish modulo p. For $p = 3$, if $n > 3j$, the factor $(-3)^{n-3j}$ is divisible by 3. If $n = 3j$ and $j > 0$, then $(3j)!/(j!^3)$ is divisible by 3, since adding $j + j + j$ in base 3 creates a carry. Thus $s_9(0) = 1$ and $s_9(n) \equiv 0 \pmod{3}$ for $n > 0$.

Multiplication by $\binom{2n}{n}$ preserves Lucas congruences, so the five $U_i(n)$ are Lucas. Finally, $(-1)^n$ is Lucas for odd primes and is invisible modulo 2, so $U_8^-(n) = (-1)^n U_8(n)$ is Lucas as well. □

This theorem is independent of the C/F bridge. It is also independent of the BTY theorem-transfer mechanism. The level-6 rows have a separate local BTY/Atkin–Lehner audit, and the V8 and V9 normalizer geometry remains useful modular context, but neither is used in the proof above. No raw weight-one $\chi(w_d)$ values are computed here.

Other source layers, including Liu's binomial-transform congruence profile, Straub's Gessel–Lucas theorem, and Beukers–Tsai–Ye's modular-form Lucas congruence source [Liu25, Str24, BTY25], are summarized in Appendix E. They are kept brief here because their role is source attachment and claim discipline, not the main C/F theorem.

9 Open-problem atlas

The open-problem atlas records blockers rather than vague directions. A selection is shown in Table 2; a fuller schema appears in Appendix D.

Branch	Blocking lemma or missing contract	Status
BTY weight-one rows	row-specific $\chi(w_d)$ values	open
Straub derivative transfer	low-carry identity for the corrected connection	proof obligation
Dwork model comparison	controlled equivalence between complement model and Q_F	open
Rowland–Yassawi source application	full model/state certificates not distributed	metadata only
Edge-watch scan	no unresolved V1 watch rows remain	closed metadata
Sun raw-C target	proof reported outside the C/F atlas	separate manuscript
A2b historical branch	no independent closure asserted here	historical metadata

Table 2: Selected open-problem and blocker rows.

10 Edge scan and watch-triage discipline

The historical C/F-like Möbius-binomial scan record reports 17684 transforms of the form

$$B(x) = (1-ax)^{-r} A\left(\frac{sx}{1-ax}\right).$$

The recorded scan used exact integer arithmetic on 12 coefficients, the 16 catalog families recorded in the companion data, $r \in \{1,2,3\}$, $-200 \le a \le 200$, and nonzero integers s with $|s| \le 20$. These are reported historical counts, not an exhaustive Cartesian enumeration of the displayed parameter ranges: the producer and eligibility trace giving that denominator are not included. The present suite recounts the twelve stored hits and checks the recorded summaries; it does not rerun the scan. The record recovered known or canonical relations and promoted no new theorem edge. The eight previous watch hits were subsequently resolved as known metadata: six are alias/canonical identity hits for the Apéry zeta(3) and Almkvist–Zudilin gamma catalog rows, and two are inverse Franel-transform hits. This is a negative discipline result: coefficient matches are not automatically promoted.

11 Data package and reproducibility

The data package contains registry rows, claim classes, source labels, proof obligations, open-problem statuses, edge-scan triage records, manifests, and readouts. The paper-facing purpose is reproducibility of claim status: a reader should be able to tell which rows are proved in this paper, which are imported by reference, which are finite replays, and which are blocked. The package is based on the V2 manifest, with the V1 registry retained as immutable provenance for

the original C/F theorem-transfer rows. Appendix E lists source identifiers and package fields. The curated companion package is available at

`https://github.com/aconsciousfractal/C-F_Bridge_and_Atlas`,

with the paper source in `paper/`, JSON artifacts in `data/relation_atlas/`, replay scripts in `scripts/`, and the executable-scope note in `docs/atlas/EXECUTABLE_REPLAY_SCOPE.md`.

12 Conclusion

We do not solve every problem attached to the C/F bridge. Instead, we give a normalized proof of the bridge and a reproducible model for using it without overclaiming. The bridge transports generating functions, coefficients, ODEs, ordinary Hankel matrices and their Smith forms, and a constant-term complement model. The atlas records adjacent theorem sources, metadata layers, finite checks, conjectures, blockers, and bounded negatives with distinct claim classes. This separation is the main lesson: bridge identities are powerful precisely when their scope is explicit.

A Theta-gauge certificate details

This appendix gives a self-contained small certificate for the theta-ratio identity used in Theorem 3.1, with the modular parametrizations imported in Section 3.1. Put

$$R = \frac{\Theta_F}{\Theta_C} = \frac{\eta_1^9 \eta_6^3}{\eta_2^9 \eta_3^3}, \qquad H = R - (1 - 9t_C).$$

The eta exponent vectors, in the order $1, 2, 3, 6$, are

$$t_C : (4, -8, -4, 8), \qquad h : (-5, 1, -1, 5), \qquad R : (9, -9, -3, 3).$$

Each has weight zero, satisfies the two eta-quotient congruences modulo 24, and has trivial character. Their cusp orders follow from the eta-quotient order formula

$$\operatorname{ord}_{a/c}\left(\prod_{\delta|6} \eta_\delta^{r_\delta}\right) = \frac{6}{24c\,\gcd(c, 6/c)} \sum_{\delta|6} \frac{\gcd(c,\delta)^2}{\delta} r_\delta, \qquad c \mid 6,$$

where $c = 6$ represents the cusp at infinity. The orders are measured in the respective local cusp parameters:

object	0	1/2	1/3	∞
t_C	0	-1	0	1
h	-1	0	0	1
R	1	-1	0	0

Eta has no zeros in the upper half-plane. Hence H is a meromorphic function on $X_0(6)$ with no interior poles and with at most one simple pole, at the cusp $1/2$.

Using $\eta(q) = q^{1/24} \prod_{n\geq 1}(1 - q^n)$, the displayed products give

$$t_C = q + O(q^2), \qquad R = 1 - 9q + O(q^2), \qquad H = O(q^2).$$

A nonzero meromorphic function on the compact curve $X_0(6)$ has equally many zeros and poles counted with multiplicity. A zero of order at least two at infinity is incompatible with a total pole order at most one. Thus $H = 0$ and

$$\Theta_F/\Theta_C = 1 - 9t_C.$$

The inverse ratio now follows algebraically:

$$\Theta_C/\Theta_F = (1 - 9t_C)^{-1} = 1 + 9h = 1 - 9t_F^{\mathrm{cat}}.$$

This proof requires only the two leading terms explicitly derived above, not a long coefficient replay or an executable Sturm certificate engine.

A.1 F twist transport dependency

The F-side parameter in source notation is twist-guarded. If $F(x) = \sum_{n\geq 0} F_n x^n$ denotes the catalog-normalized generating function, the transport contract is

$$F\big(-t_F^{\mathrm{source}}(q)\big) = \Theta_F^{\mathrm{source}}(q),$$

or equivalently

$$t_F^{\mathrm{cat}}(q) = -t_F^{\mathrm{source}}(q).$$

This is why Theorem 3.1 is stated in catalog-normalized F notation and why the inverse theta-gauge identity is written as $\Theta_C/\Theta_F = 1 - 9t_F^{\mathrm{cat}}$.

A.2 Atkin–Lehner W2 parameter and theta audit

The C/F normalization also admits a precise Atkin–Lehner description. Put

$$h = t_F^{\mathrm{source}}, \qquad t_C = \frac{h}{1+9h}, \qquad t_F^{\mathrm{cat}} = -h,$$

and let

$$W_2 = \begin{pmatrix} 2 & -1 \\ 6 & -2 \end{pmatrix}.$$

For weight-zero functions, $f|W$ means precomposition with W. For a weight-one form we use the convention

$$(f|_1 W)(\tau) = \det(W)^{1/2}(c\tau + d)^{-1} f(W\tau),$$

with the positive real branch $\det(W_2)^{1/2} = \sqrt{2}$.

Lemma A.1 (Atkin–Lehner W_2 audit)**.** *In the above normalization,*

$$t_C|W_2 = 1 + 8t_F^{\mathrm{source}} = 1 - 8t_F^{\mathrm{cat}},$$

and the theta eta quotients satisfy

$$\Theta_C|_1 W_2 = (2\sqrt{2})^{-1}\Theta_F, \qquad \Theta_F|_1 W_2 = -2\sqrt{2}\,\Theta_C.$$

Proof. The parameter identity follows by substituting the normalized action

$$W_2(h) = -\frac{h + 1/8}{9(h + 1/9)}$$

into $t_C = h/(1+9h)$. This gives $t_C|W_2 = 1 + 8h$.

For the theta statement, write $\eta_d(\tau) = \eta(d\tau)$. The eta quotients are

$$\Theta_C = \frac{\eta_2^6\eta_3}{\eta_1^3\eta_6^2}, \qquad \Theta_F = \frac{\eta_1^6\eta_6}{\eta_2^3\eta_3^2}.$$

The four eta arguments $dW_2\tau$, for $d \in \{1,2,3,6\}$, factor projectively through $\mathrm{SL}_2(\mathbb{Z})$ matrices as follows:

d	γ_d	factorization	target eta index
1	$\begin{pmatrix} 1 & -1 \\ 3 & -2 \end{pmatrix}$	$\gamma_1\ \mathrm{diag}(2,1)$	2
2	$\begin{pmatrix} 2 & -1 \\ 3 & -1 \end{pmatrix}$	$\gamma_2\ \mathrm{diag}(1,1)$	1
3	$\begin{pmatrix} 1 & -3 \\ 1 & -2 \end{pmatrix}$	$\gamma_3\ \mathrm{diag}(6,1)$	6
6	$\begin{pmatrix} 2 & -3 \\ 1 & -1 \end{pmatrix}$	$\gamma_6\ \mathrm{diag}(3,1)$	3.

Using the standard Dedekind eta multiplier formula

$$\eta(\gamma\tau) = \exp(\pi i R(\gamma))(c\tau + d)^{1/2}\eta(\tau), \qquad R(\gamma) = \frac{a+d}{12c} - s(d,c) - \frac{1}{4},$$

with the principal square root for $c > 0$, the exact exponents are

d	1	2	3	6
$R(\gamma_d)$	$-1/3$	$-1/6$	$-1/3$	$-1/6.$

For Θ_C, with eta exponent vector $(-3, 6, 1, -2)$, the total root-of-unity exponent is

$$-3(-1/3) + 6(-1/6) + 1(-1/3) - 2(-1/6) = 0.$$

For Θ_F, with exponent vector $(6, -3, -2, 1)$, it is

$$6(-1/3) - 3(-1/6) - 2(-1/3) + 1(-1/6) = -1.$$

The remaining τ-dependent factors cancel against the weight-one slash prefactor. The residual powers of 2 are $2^{-3/2}$ for Θ_C and $-2^{3/2}$ for Θ_F, giving the two displayed identities. □

This audit is a normalization statement, not a replacement for the bridge. Indeed, in the coordinate $T = t_C$,

$$T|W_2 = \frac{1-T}{1-9T}, \qquad g(T) = -\frac{T}{1-9T},$$

so the catalog C/F bridge map is not literally the Atkin–Lehner action. Rather,

$$g(T) = \frac{1-(T|W_2)}{8}.$$

Thus the F catalog parameter is an affine normalization of the W_2-translate of the C parameter. This observation also does not compute the row-specific BTY multipliers $\chi(w_d)$, which remain outside the closed claims of this paper.

B Binomial-complement constant-term lemma

Lemma 5.1 is included in the main text because it is the cleanest bridge between coefficient transforms and constant-term models. The important guardrail is repeated here: the operation

$$\Lambda \mapsto a - \Lambda$$

preserves the fact that one has a constant-term sequence, but it is not a Laurent mutation claim and does not imply Newton equivalence between two chosen Laurent models.

C Registry schema and selected rows

Each registry row uses the following schema:

`statement_id`	row identifier
`source_paper`	external source or current paper
`source_label`	the theorem, conjecture, or artifact label
`source_object`	source sequence or model
`target_object`	target sequence or model
`rewrite`	transport or normalization formula
`cf_role`	how the C/F bridge enters
`claim_level`	claim class
`finite_replay`	finite evidence, if any
`literature_status`	source status
`proof_obligation`	missing lemma, if any
`status`	current row status
`scope_note`	public limitation or remaining mathematical condition.

A representative public excerpt is:

statement_id	source	target	rewrite/status	claim level
cf_bridge	C	F	$F = \mathcal{T}(C)$	proved in paper
cf_involution	C,F	F,C	$\mathcal{T}^2 = 1$	proved in paper
cf_hankel	C,F	C,F	$\Delta_N(C) = \Delta_N(F)$	proved in paper
cf_ode_pullback	L_C	L_F	rational pullback	proved in paper
sun_f_to_tc	Sun F-side	$T(C)$	signed transfer only	conditional transfer
sun_raw_c	raw C	RV-108	no C/F transfer	separate proof obligation
dwork_ct	$\Lambda_{C,1}$	$9 - \Lambda_{C,1}$	CT complement	closed metadata
ry_automata	small moduli	automata	fixed-modulus data	closed metadata
chan_cooper				
five_companion				
lucas	Chan–Cooper	U_i rows	direct Lucas proof	proved in paper

The frozen V1 machine-readable table is the JSON file

`data/relation_atlas/`
`cf_theorem_transfer_registry_v1.json`.

The June 2026 paper-facing addendum rows, including Hankel, Atkin–Lehner W_2, and Chan–Cooper Lucas, are recorded separately in

`data/relation_atlas/`
`cf_theorem_transfer_registry_`
`2026_06_extension.json`

and summarized in the V2 registry/readout documents. The October addendum records Corollary 3.4 and a source-correction overlay. Its JSON file is listed in the public atlas index. The example identifiers above are descriptive labels, not literal V1 row IDs. The excerpt is included only to make the atlas semantics visible in the paper.

D Open problems and blocking lemmas

The atlas records open work by the mathematical object that is missing. For the Sun branch, the C/F theorem transports F-side statements to $T(C)$, but it does not itself supply the raw-C/RV-108 bridge. In the historical V1/V2 data package this appears as a deferred proof obligation. After that atlas freeze, a proof of the raw-C target is reported in the separate manuscript on Sun's supercongruence [Bab26]. We keep the registry row as provenance for the C/F atlas rather than reclassifying it as a C/F consequence. This paper does not certify that separate proof or the historical A2b branch.

The modular-form and constant-term branches have different blockers. The Beukers–Tsai–Ye weight-one rows require row-specific multiplier data $\chi(w_d)$ before they can be used as theorem imports. The Dwork branch has a proved C/F constant-term complement operation, but the stronger comparison between the complement model and the Gorodetsky F-side model Q_F remains an equivalence-or-obstruction problem. The Rowland–Yassawi branch has explicit small automata as metadata, but a complete public certificate package for those recorded automata would still require the explicit model contracts, denominator units, and the complete Cartier-state certificates. The historical generic bad-prime/singularity wording is corrected by the October source overlay.

Finally, the Straub derivative-transfer branch is not closed by the formal C/F coefficient involution alone. The corrected connection term leaves a finite low-carry identity as the remaining proof obligation. The edge-watch scan no longer contributes an open problem: its eight watch rows have been resolved as known metadata.

E Source and bibliography status

Zagier and OEIS fix the catalog-normalized C and F rows used throughout the paper. Beukers supplies the modular parametrizations, the common $\Gamma_0(6)$ setting, and the F twist caveat used in the theta-gauge certificate. Barrucand, Callan, and Verrill supply the classical Franel transform context that explains the C/F bridge as part of a known network rather than as an isolated coefficient coincidence.

The remaining sources enter as theorem imports or metadata layers. Sun supplies the source congruence layer and the original Conjecture 3.5 target [Sun22]; the latter is addressed in a separate supercongruence manuscript rather than by the C/F bridge [Bab26]. Liu supplies binomial-transform congruence metadata with convention guards [Liu25]. In the Liu audit, thirteen rows match the normalized catalog directly, while the η and $s18$ rows are stored with an explicit convention map rather than as failures. Chan–Cooper supplies the companion binomial formulas for the five rows in Theorem 8.2. Our U_i are the central-binomial lifts of the source s_i, not the distinct Chan–Cooper $t(k)$ sequences; the Lucas proof is given in this paper and does not import a BTY theorem [CC12]. Straub's Gessel–Lucas theorem is imported only under its hypotheses $p \geq 3$ and $0 \leq k < p$, including the author's corrigendum to the printed range [Str]; the separate C/F derivative-transfer problem remains guarded by its low-carry proof obligation [Str24].

The Beukers–Tsai–Ye theorem is used only when row hypotheses are complete [BTY25]. Mellit–Vlasenko and Gorodetsky provide the Dwork and constant-term context, and Rowland–Yassawi provides the source contract for automatic congruences [MV16, Gor23, RY15]. The broader tabular background of Calabi–Yau and Apéry-like operators is represented by the Almkvist–van Enckevort–van Straten–Zudilin tables and the subsequent transformation literature [AvEvSZ05, AvSZ11]. Before submission, the remaining bibliographic work is journal-specific style and final access-date policy for web catalog entries.

References

[AvEvSZ05] Gert Almkvist, Christian van Enckevort, Duco van Straten, and Wadim Zudilin. Tables of Calabi–Yau equations, 2005.

[AvSZ11] Gert Almkvist, Duco van Straten, and Wadim Zudilin. Generalizations of Clausen's formula and algebraic transformations of Calabi–Yau differential equations. *Proceedings of the Edinburgh Mathematical Society*, 54(2):273–295, 2011.

[Bab26] Oleksiy Babanskyy. Sun's supercongruence for an Apéry-like sequence, 2026. Manuscript and public companion repository.

[Bar75] P. Barrucand. A combinatorial identity. *SIAM Review*, 17:168, 1975. Problem 75-4.

[Beu25] Frits Beukers. Supercongruences using modular forms, 2025. arXiv:2403.03301v3, preprint; last revised 16 June 2025.

[BTY25] Frits Beukers, Wei-Lun Tsai, and Dongxi Ye. Lucas congruences using modular forms. *Bulletin of the London Mathematical Society*, 57(1):69–78, 2025.

[Cal08] David Callan. A combinatorial interpretation for an identity of Barrucand. *Journal of Integer Sequences*, 11(3):Article 08.3.4, 2008.

[CC12] Heng Huat Chan and Shaun Cooper. Rational analogues of Ramanujan's series for $1/\pi$. *Mathematical Proceedings of the Cambridge Philosophical Society*, 153(2):361–383, 2012.

[Gor23] Ofir Gorodetsky. New representations for all sporadic Apéry-like sequences, with applications to congruences. *Experimental Mathematics*, 32(4):641–656, 2023.

[Liu25] Ji-Cai Liu. On the binomial transforms of Apéry-like sequences. *Canadian Mathematical Bulletin*, 68(2):359–376, 2025.

[MV16] Anton Mellit and Masha Vlasenko. Dwork's congruences for the constant terms of powers of a laurent polynomial. *International Journal of Number Theory*, 12(2):313–321, 2016.

[OEI26a] OEIS Foundation Inc. Entry A002893 in The On-Line Encyclopedia of Integer Sequences. `https://oeis.org/A002893`, 2026. Accessed 10 May 2026.

[OEI26b] OEIS Foundation Inc. Entry A093388 in The On-Line Encyclopedia of Integer Sequences. `https://oeis.org/A093388`, 2026. Accessed 10 May 2026.

[RY15] Eric Rowland and Reem Yassawi. Automatic congruences for diagonals of rational functions. *Journal de Théorie des Nombres de Bordeaux*, 27(1):245–288, 2015.

[SS06] Michael Z. Spivey and Laura L. Steil. The k-binomial transforms and the Hankel transform. *Journal of Integer Sequences*, 9(1):Article 06.1.1, 2006.

[Str] Armin Straub. Corrigendum to gessel–lucas congruences for sporadic sequences. Undated author's correction: the digit range in Theorems 1.2, 1.3 and 5.1 is $0 \leq k < p$; accessed 10 October 2026.

[Str24] Armin Straub. Gessel–lucas congruences for sporadic sequences. *Monatshefte für Mathematik*, 203:883–898, 2024.

[Sun22] Zhi-Hong Sun. Supercongruences involving Apéry-like numbers and binomial coefficients. *AIMS Mathematics*, 7(2):2729–2781, 2022.

[Ver10] H. A. Verrill. Some congruences related to modular forms. *International Journal of Number Theory*, 6(6):1367–1390, 2010. MPIM Preprint Series 1999 (26).

[Zag09] Don Zagier. Integral solutions of Apéry-like recurrence equations. In *Groups and Symmetries*, volume 47 of *CRM Proceedings and Lecture Notes*, pages 349–366. American Mathematical Society, Providence, RI, 2009.